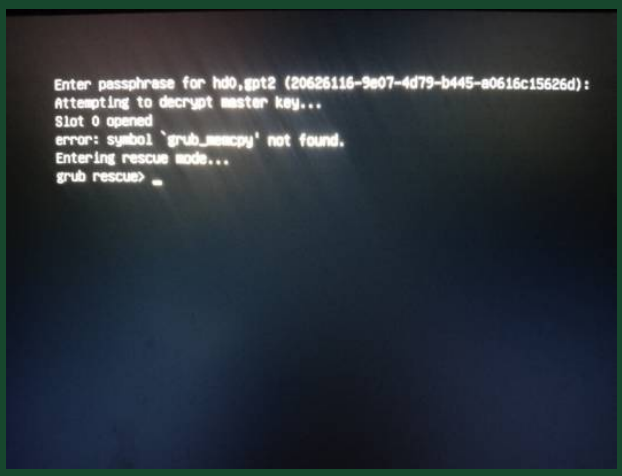
GRUB rescue fix



You're stuck in **GRUB rescue mode** because of this error:

```
error: symbol 'grub_memcpy' not found.
Entering rescue mode...
grub rescue>
```

This usually happens after:

- A **failed system update**
- A **GRUB version mismatch**
- A corrupted **/boot** partition
- Disk/partition changes

Since it's asking for a passphrase, you're using **LUKS disk encryption**, which is fine — but GRUB itself is broken.

✅ Fix Method (Recommended)

You need to **reinstall GRUB from a Linux Live USB**.

🔹 Step 1: Boot From a Linux Live USB

Use:

- Ubuntu Live USB (recommended)
- Or whatever distro you installed

Select: **"Try Ubuntu"** (or similar live mode)

🔹 Step 2: Open Terminal in Live Environment

Then check your partitions:

```
lsblk
```

Identify:

- Your root partition (likely something like `/dev/sda2`)
- Your EFI partition (something like `/dev/sda1`, usually 100–500MB, FAT32)

If you're unsure, send me the `lsblk` output.

🔹 Step 3: Unlock Encrypted Partition (IMPORTANT)

Since yours is encrypted:

```
sudo cryptsetup luksOpen /dev/sda2 cryptroot
```

Enter your passphrase.

Then:

```
sudo mount /dev/mapper/cryptroot /mnt
```

🔹 Step 4: Mount EFI Partition

```
sudo mount /dev/sda1 /mnt/boot/efi
```

(Replace with your actual EFI partition.)

🔹 Step 5: Bind System Directories

```
for i in /dev /dev/pts /proc /sys /run; do sudo mount --bind $i /mnt$i; done
```

🔹 Step 6: Chroot into Your System

```
sudo chroot /mnt
```

Now you're inside your installed system.

🔹 Step 7: Reinstall GRUB

For UEFI systems:

```
grub-install
update-grub
```

If that fails:

```
grub-install --target=x86_64-efi --efi-directory=/boot/efi
update-grub
```

🔹 Step 8: Exit and Reboot

```
exit
sudo reboot
```

Remove the USB.

🔥 Why This Happens

The `grub_memcpy` error means:

- Your **GRUB core image and modules don't match**
- Often caused by a partial upgrade or interrupted install

Reinstalling GRUB fixes the mismatch.

⚠️ If You Don't Have a Live USB

Tell me:

- Your Linux distro
- Whether this is dual boot with Windows
- Whether you updated recently

I'll give you an alternative approach.

Take your time. Don't panic — this is fixable.
Send me your `lsblk` output if you want me to guide you step-by-step.

```
 sudo cryptsetup luksDump /dev/sda2                                  ✔
LUKS header information for /dev/sda2

Version:        1
Cipher name:    aes
Cipher mode:    xts-plain64
Hash spec:      sha256
Payload offset: 4096
MK bits:        512
MK digest:      7e 12 33 f1 9d ef ac 56 b8 dd d4 e4 64 88 43 2d c6 44 35 b2
MK salt:        8d 14 14 3e 90 01 4c 57 2d 8b 28 9d 5b 36 a0 e5
                ba 4a ea 9e 17 bb d0 c0 84 a6 2d 9c 85 b9 05 06
MK iterations:  314698
UUID:           20626116-9e07-4d79-b445-a0616c15626d

Key Slot 0: ENABLED
    Iterations:         5011114
```

```
        Salt:               17 15 ee ca 4c 6c 93 4a 34 c3 28 51 41 4c 56 31
                            b6 28 b8 e6 ea d7 2e dd e8 dc f1 6f 7f 7c 30 85
        Key material offset:   8
        AF stripes:            4000
  Key Slot 1: ENABLED
        Iterations:            5029140
        Salt:               f8 41 e3 25 32 3a de c8 42 58 96 e3 bd 75 d5 a4
                            6b 04 64 90 fa 84 ab ef b6 26 a4 dd e4 f0 4a 68
        Key material offset:   512
        AF stripes:            4000
  Key Slot 2: DISABLED
  Key Slot 3: DISABLED
  Key Slot 4: DISABLED
  Key Slot 5: DISABLED
  Key Slot 6: DISABLED
  Key Slot 7: DISABLED
   ~  lsblk                                                    ✔
  NAME      MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
  loop0       7:0    0    71M  1 loop /run/miso/sfs/livefs
  loop1       7:1    0   929M  1 loop /run/miso/sfs/mhwdfs
  loop2       7:2    0   833M  1 loop /run/miso/sfs/desktopfs
  loop3       7:3    0 858.9M  1 loop /run/miso/sfs/rootfs
  sda         8:0    0 238.5G  0 disk
  ├─sda1      8:1    0   300M  0 part
  ├─sda2      8:2    0 188.2G  0 part
  └─sda3      8:3    0    50G  0 part /run/media/manjaro/New Volume
  sdb         8:16   1  29.3G  0 disk
  ├─sdb1      8:17   1  29.3G  0 part
  │ ├─ventoy  254:0    0   2.8G  1 dm   /run/miso/bootmnt
  │ └─sdb1    254:1    0  29.3G  0 dm
  └─sdb2      8:18   1    32M  0 part
  nvme0n1     259:0    0 476.9G  0 disk
  ├─nvme0n1p1 259:1    0   100M  0 part
  ├─nvme0n1p2 259:2    0    16M  0 part
  └─nvme0n1p3 259:3    0 476.8G  0 part
   ~
```

Good. You're booted into **Manjaro live USB** (I can see `/run/miso/...`), and your encrypted system is:

- `sda2` → LUKS (your Linux root)
- `sda1` → 300MB (very likely EFI)
- `sda3` → 50GB (mounted as "New Volume" — probably Windows data)

Your encrypted header is fine. Nothing is corrupted. This is almost certainly just a broken GRUB install.

Now we fix it properly.

✅ Step-by-Step Fix (Manjaro Live USB)

1️⃣ Unlock the encrypted partition

```
sudo cryptsetup luksOpen /dev/sda2 cryptroot
```

Enter your passphrase.

Then check:

```
ls /dev/mapper
```

You should see `cryptroot`.

2️⃣ Mount your system

```
sudo mount /dev/mapper/cryptroot /mnt
```

3️⃣ Mount EFI partition

```
sudo mount /dev/sda1 /mnt/boot/efi
```

If that errors, create the directory first:

```
sudo mkdir -p /mnt/boot/efi
sudo mount /dev/sda1 /mnt/boot/efi
```

4️⃣ Use Manjaro's built-in chroot tool (IMPORTANT)

Since you're using Manjaro, do NOT use manual chroot.

Use:

```
sudo manjaro-chroot /mnt
```

This handles everything automatically.

5️⃣ Reinstall GRUB (UEFI system)

Inside chroot:

```
grub-install --target=x86_64-efi --efi-directory=/boot/efi --bootloader-id=manjaro
update-grub
```

If you get no errors, you're good.

6️⃣ Exit and reboot

```
exit
sudo reboot
```

Remove USB.

🔥 Why This Happened

That error:

```
error: symbol 'grub_memcpy' not found
```

Means:

GRUB core image ≠ GRUB modules version

Usually caused by:

- Interrupted update
- Kernel/GRUB update without reinstall
- Manjaro update not fully completed

Reinstalling GRUB fixes the mismatch.

⚠️ Important Question

Your `lsblk` shows another disk:

```
nvme0n1 (476GB)
```

Are you sure your Linux is on `sda2` and not on NVMe?

If this was a laptop upgrade or disk change, that could matter.

Before rebooting, tell me:

- Is this a dual boot with Windows?
- Was there a recent update before this broke?

We're very close to fixing this permanently.

```
 sudo cryptsetup luksOpen /dev/sda2 cryptroot                    ✔
Enter passphrase for /dev/sda2:
  ~  ls /dev/mapper                                    ✔  6s
control  cryptroot  sdb1  ventoy
  ~  sudo mount /dev/mapper/cryptroot /mnt                    ✔
  ~  sudo mkdir -p /mnt/boot/efi                    ✔
sudo mount /dev/sda1 /mnt/boot/efi
  ~  sudo manjaro-chroot /mnt                    ✔
sh-5.3# grub-install --target=x86_64-efi --efi-directory=/boot/efi --bootloader-id=manjaro
update-grub
Installing for x86_64-efi platform.
Installation finished. No error reported.
Generating grub configuration file ...
Found theme: /usr/share/grub/themes/manjaro/theme.txt
Found linux image: /boot/vmlinuz-6.12-x86_64
Found initrd image: /boot/intel-ucode.img /boot/amd-ucode.img /boot/initramfs-6.12-x86_64.img
Found linux image: /boot/vmlinuz-5.10-x86_64
Found initrd image: /boot/intel-ucode.img /boot/amd-ucode.img /boot/initramfs-5.10-x86_64.img
```

```
Warning: os-prober will be executed to detect other bootable partitions.
Its output will be used to detect bootable binaries on them and create new boot entries.
ERROR: mkdir /var/lock/dmraid
Adding boot menu entry for UEFI Firmware Settings ...
Root filesystem isn't btrfs
If you think an error has occurred, please file a bug report at "https://github.com/Antynea/grub-btrfs"
Found memtest86+ image: /boot/memtest86+/memtest.bin
Found memtest86+ EFI image: /boot/memtest86+/memtest.efi
done
sh-5.3#
```

Perfect ✅ — this is exactly what we wanted to see.

✔️ What just happened

- `grub-install` → **No error reported** ✅
- `update-grub` → Detected both kernels (6.12 and 5.10) ✅
- EFI entry created ✅

The only message:

```
ERROR: mkdir /var/lock/dmraid
```

is harmless in this situation. It does **NOT** affect boot.

Everything important succeeded.

🔥 Now Do This

Inside chroot, type:

```
exit
```

Then:

```
sudo reboot
```

Remove the USB when it starts rebooting.

🚨 If It Still Boots Into Rescue Mode

Then the issue is likely:

- BIOS boot order changed
- Wrong EFI entry is first
- Another disk (NVMe) is being selected

If that happens, go into BIOS and:

- Set **Manjaro** or the disk containing `/dev/sda` as first boot device
- Or choose the "manjaro" UEFI entry manually

⚠️ Important Observation

You have:

- `sda` (238GB) → Linux
- `nvme0n1` (476GB) → Probably Windows

If BIOS is booting from NVMe first, that could explain the GRUB rescue issue.

Go ahead and reboot now.

If it fails again, send me:

- What exactly appears on screen
- A photo if possible

We're very close.

> write me a shell script to do all these

Alright — here's a **clean, safe shell script** that automates the full GRUB repair process for your setup:

- Encrypted root: `/dev/sda2`
- EFI partition: `/dev/sda1`
- Distro: Manjaro
- UEFI system

🛠️ `fix-grub-luks.sh`

Save this as `fix-grub-luks.sh` on your Live USB system.

```bash
#!/bin/bash

set -e

# ===== CONFIGURATION =====
LUKS_PART="/dev/sda2"
EFI_PART="/dev/sda1"
MAPPER_NAME="cryptroot"
MOUNT_POINT="/mnt"
BOOTLOADER_ID="manjaro"

echo "===== GRUB Repair Script (LUKS + UEFI) ====="
echo

# Check if running as root
if [[ $EUID -ne 0 ]]; then
   echo "Please run as root: sudo ./fix-grub-luks.sh"
   exit 1
fi

echo "[1/6] Unlocking LUKS partition..."
cryptsetup luksOpen "$LUKS_PART" "$MAPPER_NAME"

echo "[2/6] Mounting root filesystem..."
mount /dev/mapper/$MAPPER_NAME "$MOUNT_POINT"

echo "[3/6] Mounting EFI partition..."
mkdir -p "$MOUNT_POINT/boot/efi"
mount "$EFI_PART" "$MOUNT_POINT/boot/efi"

echo "[4/6] Entering Manjaro chroot..."
manjaro-chroot "$MOUNT_POINT" /bin/bash <<EOF

echo "[5/6] Reinstalling GRUB..."
grub-install --target=x86_64-efi --efi-directory=/boot/efi --bootloader-id=$BOOTLOADER_ID

echo "[6/6] Updating GRUB configuration..."
update-grub

echo
echo "GRUB reinstall complete."

EOF

echo
echo "Unmounting partitions..."
umount -R "$MOUNT_POINT" || true
cryptsetup close "$MAPPER_NAME" || true

echo
echo "===== DONE ====="
echo "Now reboot and remove the USB."
```

🚀 How To Use

1️⃣ Boot into Manjaro Live USB
2️⃣ Save the script:

```bash
nano fix-grub-luks.sh
```

Paste the script → Save (Ctrl+O) → Exit (Ctrl+X)

3️⃣ Make executable:

```bash
chmod +x fix-grub-luks.sh
```

4️⃣ Run it:

```bash
sudo ./fix-grub-luks.sh
```

Enter your LUKS password when asked.

⚠️ If Your Partitions Ever Change

Just modify these lines at the top:

```
LUKS_PART="/dev/sda2"
EFI_PART="/dev/sda1"
```

If you'd like, I can also write a **more advanced version** that:

- Automatically detects EFI partition
- Automatically detects LUKS root
- Works on any disk (sda/nvme/etc)

Just tell me.